

Jaden Isaac

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Education

Amity University Bangalore (Amity School of Engineering and Technology)
B.Tech, Artificial Intelligence & Machine Learning | CGPA: 7.83/10

Expected July 2028

Projects

C++ Neural Network to WebAssembly | C++, WebAssembly, Emscripten, JavaScript

github.com/jadenisaac2005/cpp-neural-network | Jul 2025

- Reached 97.43% on the 10,000-image MNIST test set (native C++) and 97.33% through the live demo's own preprocessing and WebAssembly code, with backpropagation and dense layers written from scratch
- Shrank weights from 1.16 MB text to 398 KB Float32 binary with identical predictions on all 10,000 test images
- Fixed a silent failure where the demo ran an untrained network: the loader now verifies all 101,770 parameters loaded and disables prediction otherwise

Fake News Classifier | Python, scikit-learn, pandas, NLTK, NLP

github.com/jadenisaac2005/news-headline-classifier | Mar 2026

- Achieved 95.22% test accuracy on deduplicated WELFake articles (80/20 split, seed 42) with TF-IDF and logistic regression, vs 88.91% title-only and 86.27% Naive Bayes baselines
- Found 2,656 test articles (18.4%) with an identical copy in training (leaky-split accuracy 95.76%); deduplicating removed 8,577 surplus copies (72,134 to 63,557 articles)
- Showed the model's top-weighted features are mostly outlet style (bylines, naming conventions), not claim content; removing 38 such source/format tokens cut accuracy only to 93.54%

NumPy Neural Network from Scratch | Python, NumPy

github.com/jadenisaac2005/nn-from-scratch | Mar 2026

- Implemented SGD, Momentum and Adam by hand in NumPy (no autograd) and compared them on MNIST: 26, 4 and 2 epochs to reach 97% test accuracy; final 97.90%, 98.07%, 97.91% (100 epochs, one seeded run each)
- Diagnosed from the training curves that momentum had been implemented in EMA form (behaving like SGD); fixed it to classical momentum and added a unit test that fails on the old code

Rockfall Prediction System | Python, XGBoost, scikit-learn, FastAPI, React, Twilio

github.com/jadenisaac2005/rockfall-prediction | Sep 2025

- Served a SMOTE + StandardScaler + XGBoost risk classifier through FastAPI with three risk levels and Twilio SMS alerts at the critical level, sole developer on a 3-person team for Smart India Hackathon 2025
- Quantified the alert trade-off at the served cutoff: 98.3% precision (vs a 64.75% base rate) at 17.4% recall; ROC-AUC 0.75 on a held-out 60/20/20 split of synthetic data
- Caught and fixed threshold tuning on the test set and a mismatch between the evaluated and served model

Skills

Languages: Python, C++, TypeScript, JavaScript

ML & Data: NumPy, scikit-learn, XGBoost, pandas, NLTK, NLP

Systems & Web: WebAssembly, Emscripten, FastAPI, React, Next.js, Tailwind CSS

Tools: Git, Vercel